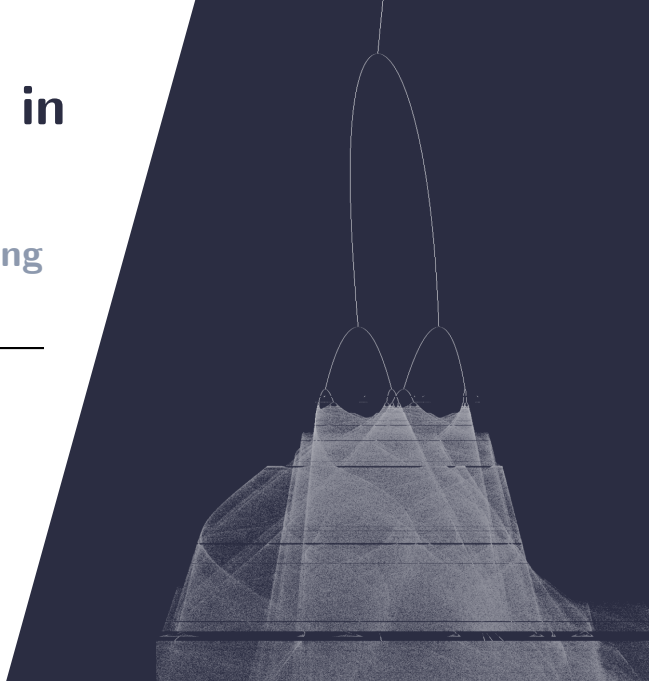
Optical Turbulence in the Ikeda Map

High-Performance Computing and Chaos Theory

Rahul Nivash

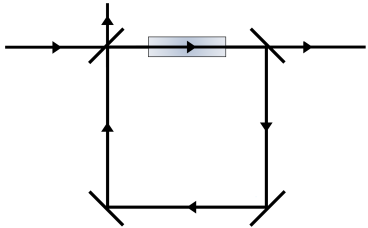
University of Manitoba

Theory Canada • June 20, 2026



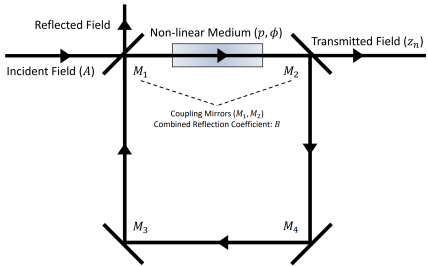
Part 1

Introduction



The Nonlinear Optical Ring Cavity

Physical Dynamics



Unidirectional nonlinear ring cavity.

- Light circulating within a non-linear ring cavity can exhibit chaotic fluctuations, a phenomenon known as optical turbulence.
- The electromagnetic field experiences feedback at distinct intervals upon reflecting off the cavity mirrors.
- This feedback requires an iterative map to capture the dynamical instabilities.

The Nonlinear Optical Ring Cavity

Historical Context

Conventional CW Theories

- Standard for modeling optical bistability prior to 1979.
- Assumed continuous phase space evolution.
- Failed to capture dynamical instabilities arising from feedback delay.

Ikeda's Prediction (1979)

- Proposed discrete-time iterative maps; i.e. $z_{n+1} = f(z_n)$.
- Predicted chaotic fluctuations (optical turbulence) due to round-trip delay.
- Shifted focus from stationary states to dynamic evolution.

The Nonlinear Optical Ring Cavity

The Ikeda Map

$$z_{n+1} = A + Bz_n \exp\left[i\left(\phi - \frac{p}{1+|z_n|^2}\right)\right]$$

Definition of Parameters:

- $z_n \in \mathbb{C}$: Transmitted electric field at the n-th iteration.
- A: Amplitude of the input light injected into the ring cavity.
- B: Reflection coefficient of the cavity mirrors.
- ϕ: Linear phase shift.
- p: Nonlinear phase shift.

Part 2

HPC Architecture



Computational Bottlenecks

The Cost of Simulating Chaos

Modeling optical turbulence requires massive computational resources:

1. The Iterative Burden

- **Complex Arithmetic:** Every iteration evaluates a complex exponential of a field-dependent phase.
- **High-Resolution Sweeps:** To observe the route to chaos, the simulation evaluates thousands to millions of reflection coefficients (B), each for thousands of round trips.

8,000 B-Parameter Steps × **5,000** Iterations per Step = **40,000,000** Map Iterations per Diagram

Computational Bottlenecks

The Cost of Simulating Chaos

Modeling optical turbulence requires massive computational resources:

2. Quantification Overhead

- **Lyapunov Exponents:** Confirming chaos requires calculating the 2×2 Jacobian matrix at every iteration step.
- **Fractal Dimensions:** Quantifying geometry of the attractor requires box-counting algorithms to repeatedly scan the entire phase space.
- **Stability Extraction:** Locating limits of the bifurcation cascade requires multi-dimensional root-finding methods.

Baseline Simulation

The Complex Iteration Algorithm

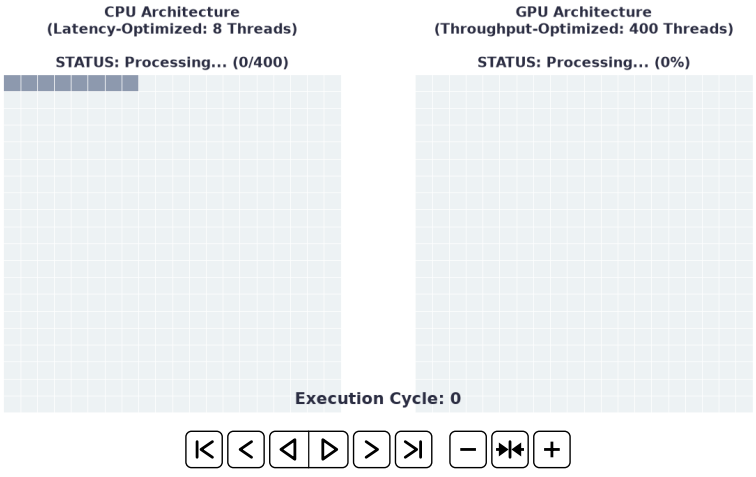
Algorithm Complex Ikeda Map Simulation

Require: **Input:** Target bounds ($B_{\text{start}}, B_{\text{end}}, B_{\text{steps}}$)
Require: **Parameters:** Input amplitude A, phase shift ϕ, nonlinear parameter p
Require: **Lengths:** $N_{\text{transient}}$ (discard phase), N_{record} (steady-state)
Ensure: **Output:** Array of asymptotic steady-state amplitudes $|z|$ mapped to B

for all $B \in B_values$ **in parallel across GPU threads do**
    $z \leftarrow 0 + 0i$ ▷ Initial condition
    for $j \leftarrow 1$ to $(N_{transient} + N_{record})$ **do**
        $z \leftarrow A + B \cdot z \cdot \exp\left[i\left(\phi - \frac{p}{1+|z|^2}\right)\right]$
        if $j > N_{transient}$ **then**
            Save $|z|$ mapped to thread index B
        end if
    end for
end for

High-Performance Computing

CPU vs. GPU Architectures



2D Decoupled Architecture

Optimizing the Ikeda Map

Original Complex Map

$$z_{n+1} = A + Bz_n \exp\left[i\left(\phi - \frac{p}{1+|z_n|^2}\right)\right]$$

2D Real-Valued Map

$$\begin{aligned} x_{n+1} &= A + B(x_n \cos t_n - y_n \sin t_n) \\ y_{n+1} &= B(x_n \sin t_n + y_n \cos t_n) \\ t_n &= \phi - \frac{p}{x_n^2 + y_n^2 + 1} \end{aligned}$$

- **No Complex Arithmetic:** One `sincos` and a few multiply-adds per round trip.
- **Closed-Form Jacobian:** Lyapunov exponents for $\sim$10 extra FLOPs per step; $\det J = B^2$.
- **One Thread per B:** Independent orbits, no locks, and coalesced memory writes.

Optimized Simulation

The 2D Decoupled Algorithm

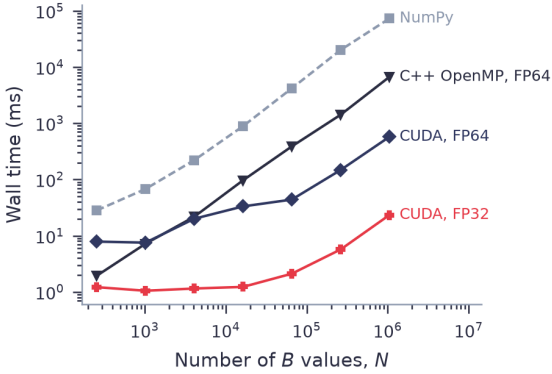
Algorithm Decoupled Ikeda Map Simulation

Require: **Input:** Target bounds ($B_{\text{start}}, B_{\text{end}}, B_{\text{steps}}$)
Require: **Parameters:** Input amplitude A, phase shift ϕ, nonlinear parameter p
Require: **Lengths:** $N_{\text{transient}}$ (discard phase), N_{record} (steady-state)
Ensure: **Output:** Arrays of steady-state coordinates (x, y) and amplitudes mapped to B

for all $B \in B_values$ **in parallel across GPU threads do**
    $x \leftarrow 0, y \leftarrow 0$ ▷ Initial condition
    for $j \leftarrow 1$ to $(N_{transient} + N_{record})$ **do**
        $t_n \leftarrow \phi - \frac{p}{x^2+y^2+1}$
        $x_{new} \leftarrow A + B(x \cos t_n - y \sin t_n)$
        $y_{new} \leftarrow B(x \sin t_n + y \cos t_n)$
        $x \leftarrow x_{new}, y \leftarrow y_{new}$
        if $j > N_{transient}$ **then**
            Save x, y, and $\sqrt{x^2 + y^2}$ mapped to thread index B
        end if
    end for
end for

Measured Performance

CUDA vs. Multithreaded C++



1000 round trips per B, Lyapunov exponent included

2^{20} values of B

C++, 12 threads, FP64	6.65 s
CUDA, FP64	0.57 s
CUDA, FP32	0.023 s
Speedup, FP64	**11.6**×
Speedup, FP32	**215**×

Quadro RTX 5000 Max-Q vs. Core i7-9750H. Speedups compare equal precision. GPU times include allocation and transfers.

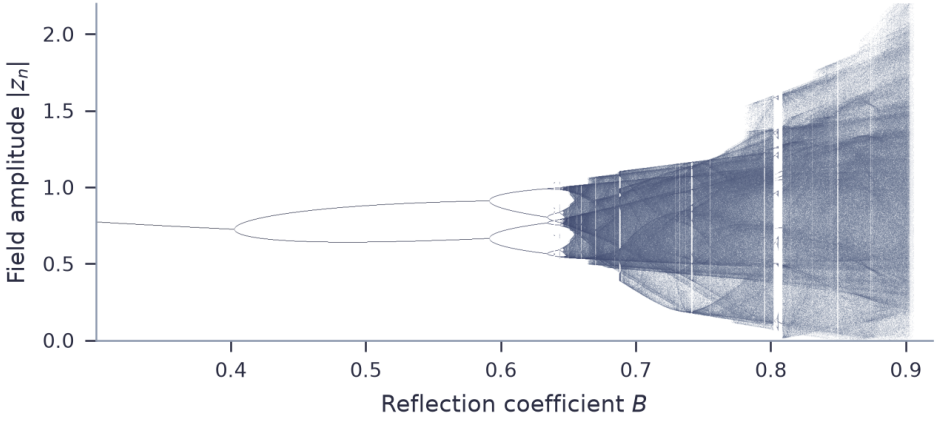
Part 3

Chaos



The Bifurcation Cascade

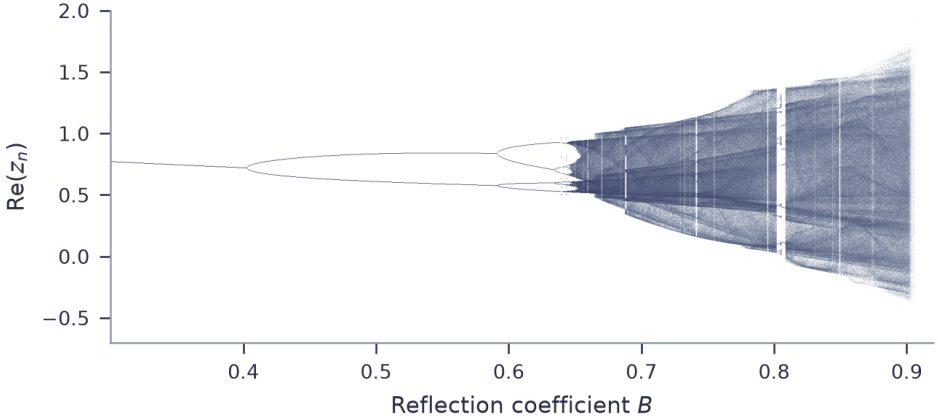
Electric Field Amplitude ($|z_n|$)



8,000 values of B. White stripes are periodic windows, e.g. period 5 at $B \approx 0.805$.

The Bifurcation Cascade

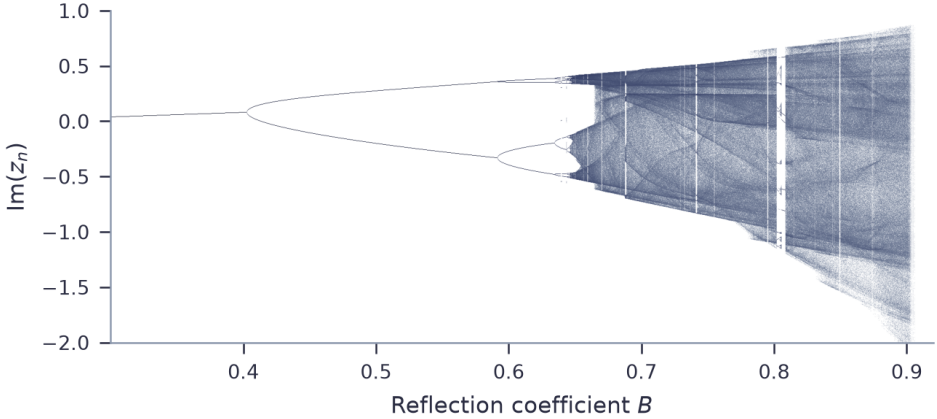
Real Component ($\mathrm{Re}(z_n)$)



Evolution of the real component $\mathrm{Re}(z_n)$ as the reflection coefficient B increases.

The Bifurcation Cascade

Imaginary Component ($\mathrm{Im}(z_n)$)



Evolution of the imaginary component $\mathrm{Im}(z_n)$ as the reflection coefficient B increases.

Quantifying Optical Turbulence

The Lyapunov Exponent (λ)

1. Theoretical Definition

The Lyapunov exponent measures the rate of divergence between two nearby trajectories in phase space.

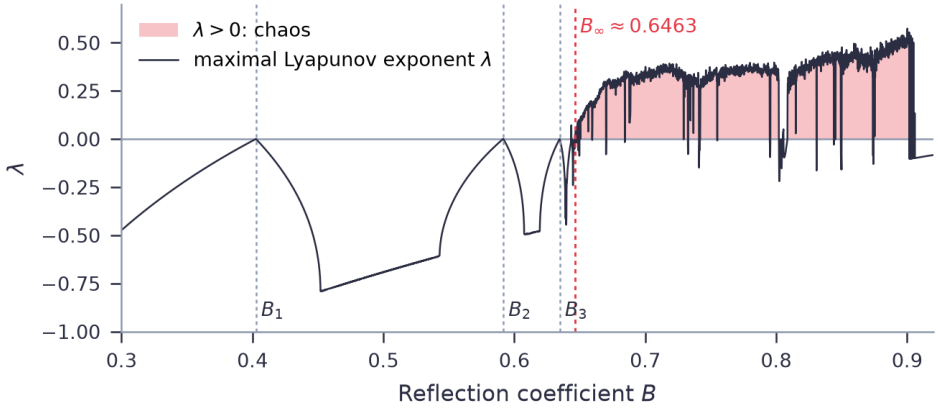
- **Sensitivity:** $|\delta z_n| \approx e^{\lambda n}|\delta z_0|$
- **Stable State:** $\lambda < 0$ (Predictable orbit)
- **Chaotic State:** $\lambda > 0$ (Optical Turbulence)

2. Computational Extraction

- **Local Linearization:** Evaluate $J_n = J(x_n, y_n)$ at every iteration step.
- **Vector Evolution:** Push a unit tangent vector forward and renormalise: $\hat{\mathbf{v}}_{n+1} = J_n\hat{\mathbf{v}}_n / \|J_n\hat{\mathbf{v}}_n\|$.
- **Asymptotic Average:** $\lambda = \lim_{N\to\infty} \frac{1}{N}\sum_{n=1}^{N} \ln \|J_n\hat{\mathbf{v}}_n\|$

Quantifying Optical Turbulence

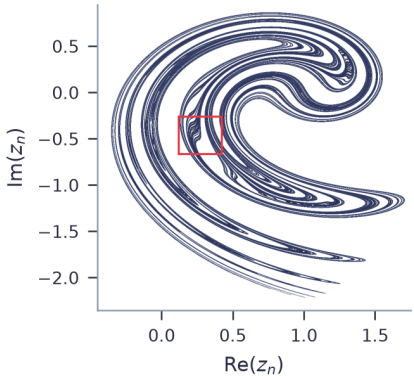
Lyapunov Spectrum Output



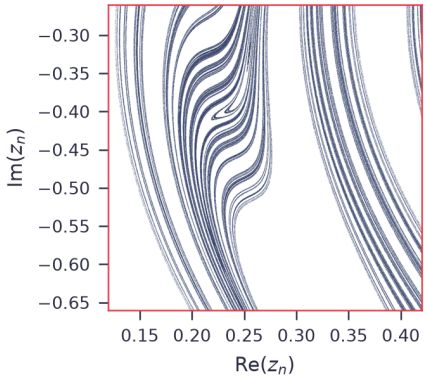
Maximal Lyapunov exponent. λ touches zero at each bifurcation B_k and turns positive (shaded) past the accumulation point $B_\infty \approx 0.6463$, apart from periodic windows.

Phase Space Geometry

The Strange Attractor



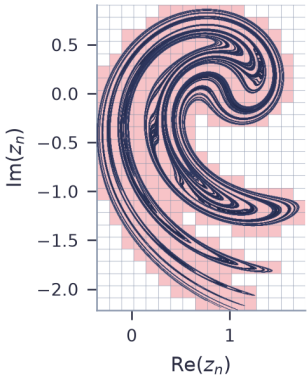
Global attractor, $B = 0.9$ (20 million round trips)



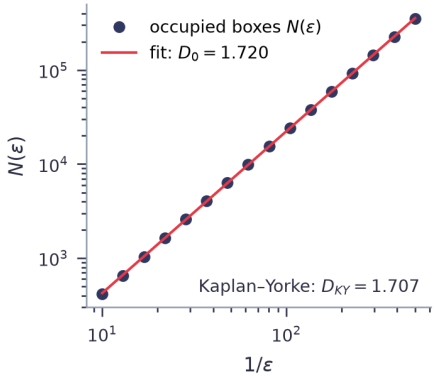
Zoomed region (red box): layered fractal folds

Quantifying Optical Turbulence

Fractal Dimension via Box-Counting



Boxes of side $\epsilon = 0.125$*:* $N(\epsilon) = 284$ *occupied (shaded)*



$D_0 = \lim_{\epsilon \to 0} \ln N(\epsilon) / \ln(1/\epsilon) = 1.720$

Numerical Results

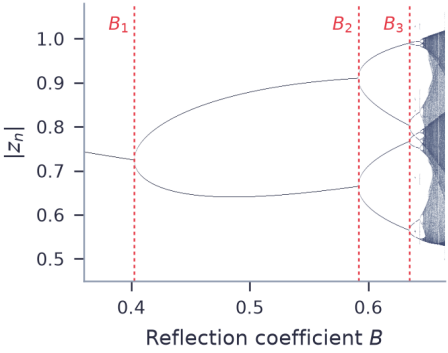
Bifurcation Thresholds and Universality

Computed Thresholds

- **Period-2 (B_1):** 0.402701
- **Period-4 (B_2):** 0.591624
- **Period-8 (B_3):** 0.634677
- **Period-16 (B_4):** 0.643845

Feigenbaum Validation:

Metric	Num.	Theo.
Ratio (δ_4)	4.666	4.669
Agreement	99.93%	



Period-doubling (flip) bifurcations at the thresholds found by Newton continuation

Theoretical Discrepancies

Continuous Steady-State vs. Discrete Delay

The Breakdown of CW Theory

- **Classical CW Models:** Assumes continuous, steady-state evolution. Predicts bistable jumps but misses chaotic dynamics.
- **The Discrete Reality:** Ikeda demonstrated that incorporating the finite round-trip delay necessitates a discrete mapping.

Threshold Discrepancy

- **CW Bistability (B_{cw}):** 0.708376
- **Instability Onset (B_1):** 0.402701
- **Onset of Chaos (B_∞):** ≈ 0.6463

Finding:
The cavity destabilises at B_1 and is chaotic by B_∞, both well before CW theory predicts any instability.

Conclusion

Summary of Findings

Research Highlights

- **HPC Optimization:** A 2D real-valued map with one GPU thread per B runs $11.6\times$ (FP64) to $215\times$ (FP32) faster than 12-thread C++.
- **Quantifying Chaos:** Jacobian-derived Lyapunov exponents and box counting confirm chaos past $B_\infty \approx 0.6463$; at $B = 0.9$, $\lambda_1 = 0.5075$ and $D_{KY} = 1.707$ match published values.
- **Theoretical Limits:** The delayed map destabilises at $B_1 = 0.4027$, far below the CW bistability threshold $B_{cw} = 0.7084$, and its cascade reproduces Feigenbaum's δ to 0.1%.

Questions?

Code, tests and benchmarks: `github.com/0xPARZE/ikeda-cuda`